A 0.9pj/b 9.8-113Gb/s XSR SerDes with 6-tap TX FFE and AC coupling RX in 3nm FinFet Technology

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Abstract

This paper presents an OIF CEI-1125G-XSR compliant short-reach SerDes with data rate from 9.8 to 113Gb/s PAM4/NRZ, implemented in 3nm finfet technology. The quarter-rate source-series terminated (SST) transmitter is implemented with 2UI-delay based MUX, supply adjusted and impedance calibrated DAC driver to maintain constant TX swing across PVT. The receiver utilizes a unique AC-coupling-based method before and after single-stage CTLE with 4 edge and 12 data samplers. The design has a standard 3 taps FFE and an optional 6 taps FFE, as well as continuous adapted RX CTLE to track temperature drift and achieve BER<1e-14 below 8dB and <1e-12 for loss up to 10dB on both 2D and advanced 2.5D CoWoS package.

Keywords: Short-Reach, Die-to-Die, SerDes, XSR, 113Gb/s, OIF, CoWoS, interposer and 3nm.

Introduction

To meet the growing demands of generative AI applications, there is an increasing requirement on cloud data center to have high bandwidth and low power inter-chip connectivity with advanced packaging. To address the low power demand, in this paper a 0.9pJ/bit transceiver architecture is proposed to operate up to 113Gb/s over a 10dB channel with low BER on both 2D and 2.5D CoWoS Package.

Circuit Design

PHY_TOP: A proposed transceiver block diagram is shown in Fig. 1, the PHY is configured as 8 TX and RX lanes with a power efficient, low jitter LC-PLL distributing the clocks to all the transceiver lanes. The TX and RX front end consists of a distributed Tcoil between BUMP and the ESD to enhance the bandwidth. RX termination can be configured as both AC and DC coupled link. In RX, 8 phase quarter- rate clock is used to sample the data and the edge. The phase of the clock is adjusted by a PI which is controlled by Bang-Bang CDR logic implemented in digital with low latency and adjustable bandwidth for optimum jitter tolerance. Inside TX, a 4-phase clock is used to generate the timing for 2UI shifter and 4:1 mux.

During startup both TX and RX, DC impedance are calibrated to match the characteristics impedance of the compliance channel which is typically 50Ω. PLL frequency calibration and duty cycle correction (DCC), TX and RX clock path DCC and DLLs are calibrated once during startup, and also runs background calibration continuously in mission mode to track any voltage or temperature drift. TX FFE coefficients can be controlled by the receiver through a Serial Packet Interface (or backchannel) to minimize residual ISI at pre1 and post 1 cursor using DSP algorithm. TX FFE coefficients can be optionally dynamically adjusted if such SPI interface is available. In RX the CTLE peaking, and DC gains are adjusted dynamically to track temperature drift and maintain optimum BER performance.

TX: The proposed quarter rate TX architecture consisting of a DAC and 2UI shifter is presented in Fig. 2. In PAM4 mode TX receives 32-bit MSB and LSB words. TX FIR combines the input data with the coefficient weight to generate 32x8 bit vector. The data path includes a 32:8x8 and 8:4x8, 2 step serialization followed by a 2UI shifter. The 2UI shifter eliminates the need of 1UI pulse generation in the clock path and thus reduces power. TX uses a 14GHz differential clock from a LC-PLL and generates 4 phase quadrature clocks using a DLL. This quarter rate clocks provides the timing for the 4:1 MUX which consists of a 4:2 serializer and a 2:1 pre driver logic. The serialization is achieved by combining the output of the 2UI shifter using a high-speed mux operating with in-phase clock CKI_4T. The final 2:1 is done using an integrated NAND/NOR logic with the Driver using quadrature phase clock CKQ_4T. The driver is implemented as 7.5bit SST DAC which provides 206 discrete levels. The driver supply is regulated and calibrated for PVT variation to maintain constant minimum required TX swing, thus reduce power.

RX: As shown in Fig. 3, the Samplers-based RX consists of three main blocks: CTLE, Samplers and Clk generator. One single CTLE stage is implemented to achieve 6dB peaking with maximum 0.2dB step. There are 20 total samplers, each of the quarter-rate path contains 3 data samplers, 1 edge sampler and 1 spare sampler. The output data are level shifted through the AC-couple circuit by level 0 for edge, +2, 0, -2 for data and ±3, ±1 for spare sampler. The sampler outputs are then deserialized by a 4:32 DEMUX which is sent to the digital. The general sampler [1][3][4] input sensitivity performance varies because the large VREF branch is added into sampler to generate large PAM4 slicing level. AC-coupled leveling method releases the large VREF branch from the sampler, thus, the sampler input sensitivity is improved. The spare sampler during mission mode allows for the data/edge samplers to be swapped with the spare sampler, thus taken offline for periodic continuous offset calibration. The sampler architecture shown in Fig. 3 uses a single phase clk with small loading cap compared to strong-arm architecture [3], thus reduces power. As there is no DFE feedback, the sampler's CK2Q is not a concern. It allows the sampler to be built with 3 cascade stages. In each stage, it has only 3 stacked devices rather than the Strong-Arm architecture [3] which need 4 stacked devices. Less stacked devices make the sampler good at working under low core supply of 0.65V. The RX clock path consists of 2 DLLs, 1 PI and buffer stages to drive the samplers, as presented in Fig. 4. The DLL1 generates 8-phase equal-spaced output clocks, from a differential cml input clock from PLL. Current-starved inverter is used as basic delay cell. Inside DLL1, delay cells are divided into 3 control groups with different control voltages (VBP1/VBN1 ~ VBP3/VBN3), hence the phase spacing between 0°-45°, 45°-

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90° and 90°-135° can be separately calibrated. This minimizes the impact of random device mismatch. DLL1 is followed by an CML Phase-Interpolator, designed for high linearity, INL/DNL <±0.5 LSB. The PI output clock is fed into DLL2. In DLL2, 0°, 90°, 180° and 270° are data sampler clk while 45°, 135°, 225° and 315° are edge sampler clk. In comparison with DLL1, the proposed DLL2 structure offers larger tuning range of data vs edge clk. This is achieved by differential tuning VBP_D2E_A/VBN_D2E_A and VBP_D2E_B/VBN_D2E_B control voltages and let the delays of corresponding delay cells change in opposite direction. A corner-tracking regulation (VDLL_REG) is used to powering both DLL1 and DLL2. By injecting a current into a replica delay cell, the generated VDLL can first-order track the corner and temperature and reduce DLL1 and DLL2 delay variations.

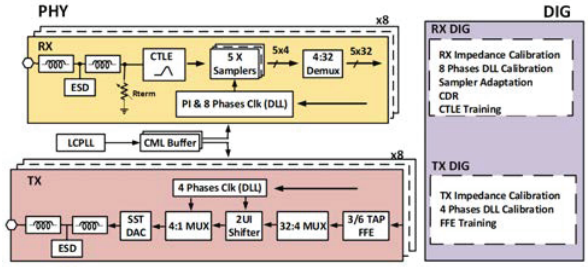


Fig. 1 113Gb/s XSR Transceiver Block Diagram.

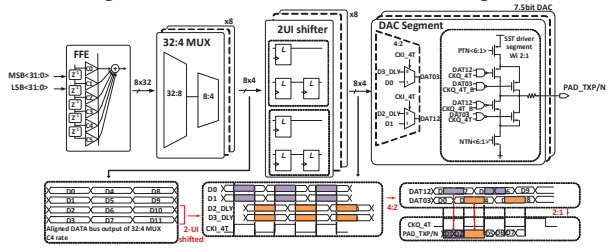


Fig. 2 TX block diagram (TOP) with timing diagram (Bottom).

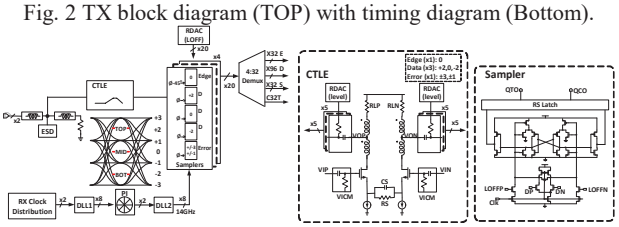


Fig. 3 RX block diagram (left), AC coupling CTLE with DAC (center) and Sampler (right).

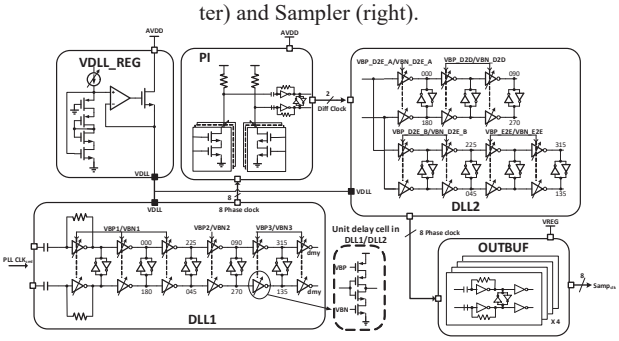
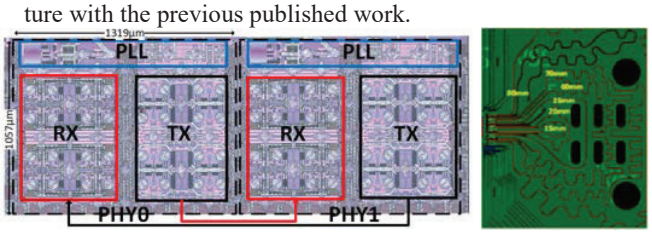


Fig. 4 RX Clock top generation block diagram with regulation.

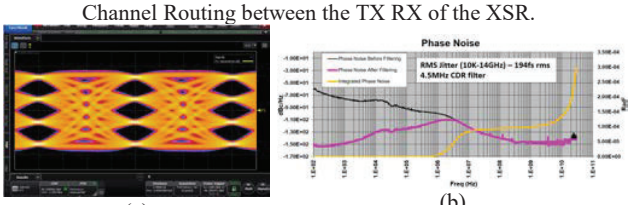
Measurement Result

A single XSR PHY consists of 8 lanes and a PLL, measures 1319µm x 1057µm area. There are 2 XSR PHYs in the test chip with PHY0 TXs are connected to PHY1 RXs and vice versa. To evaluate the performance across a wide range, a 2D and 2.5D CoWoS packages were designed with different channel lengths, between 1-80mm, which corresponds to 0.5 to 10dB IL, NEXT=-40dB and FEXT=-38dB at 28GHz. Short Channel routing and vias were optimized to reduce the reflection, impedance discontinuity and ripples in the IL and meet a COM of 2.5dB at 1E-6 BER. Fig. 5(a). shows the Die micrograph and Fig. 5(b). shows the channel routing in the package. PRSB31 data pattern is used for all measurements. Fig. 6(a) shows transmitter 112.5Gb/s PAM4 eye diagram at the bump after de_embedding and FFE equalization. Fig. 6(b) captures the phase noise of the PLL+TX. The RX Jitter Tolerance across voltage and temperature is shown in Fig. 7(a) and link BER performance over different channels in 2D and 2.5D CoWoS package at112Gb/s PAM4 is shown in Fig. 7(b). The PHY can achieve BER of <1e-14 for up to 8dB and <1e-12 for up to 10dB, with energy efficiency of 0.9pJ/bit. Table 1 compares the performance of this proposed architecture with the previous published work.



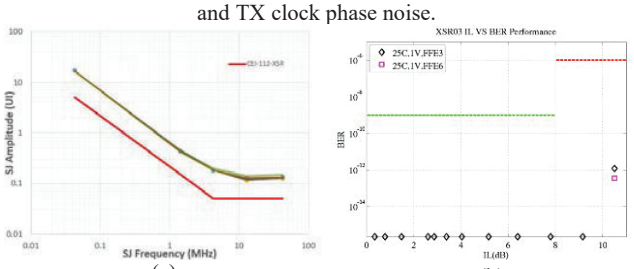
(a) (b)

Fig. 5(a) Micrograph of test chip with 2 XSR PHY (b) 2D Package Channel Routing between the TX RX of the XSR.



(a) (b)

Fig. 6 (a) Measured transmitter 112Gb/s PAM4 eye diagram (b) PLL and TX clock phase noise.



(a) (b)

Fig. 7 (a) Measured RX jitter Tolerance across PVT (b) transceiver BERs over different length package channels at112Gb/s PAM4.

Table 1 Comparison to prior work.

	This Work	ISSCC 2022 [1]	VLSI 2021 [5]	ISSCC 2021 [2]	ISSCC 2021 [3]
Technology	3NM	5NM	7NM	7NM	7NM
Data Rate(Gb/s)	113	113	112	106.25	112
Ch. Length (mm)	1 to 80	5 to 80	10 to 30	5 to 50	5 to 50
IL @ 28GHz (dB)	10.8	11.5	NA	9.6	7
Energy Eff (pJ/Bit)	0.90	1.55	1.24	1.55	1.7
Area(mm^2/lane)	0.174	0.264	0.15	0.225	0.228

References

[1] G. Gangasani et al., ISSCC 2022 [2] R. Shivnaraine et al., ISSCC 2021 [3] R. Yousry et al., ISSCC 2021 [4] A. Cevrero et al., ISSCC 2019 [5] C. Poon et al., 2021 Symp. VLSI Circuits

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